FAM: A Self-Supervised Forecast-Anything Model for Rare Event Prediction

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Abstract

Rare events in multivariate time series are defined by label scarcity: the events most critical to predict are the ones with the fewest examples. We introduce FAM, a self-supervised encoder built on a single architectural idea: *causal latent trajectory prediction* via a Joint Embedding Predictive Architecture (JEPA), which predicts future representations rather than raw values with a variable horizon $k \sim \mathcal{U}[5, 30]$. A single pretrained encoder, frozen, supports downstream event types through trivial linear probes. On C-MAPSS FD001 specifically, FAM achieves *variance-reduced* RUL prediction at low label budgets (at 5% labels: frozen 21.5 ± 0.9 vs. supervised STAR 24.6 ± 6.4; Welch's unpaired t $p = 0.36$, means statistically equivalent), with $\sim 7\times$ lower seed variance than STAR. The empirically supported finding is reliability, not mean outperformance. A from-scratch ablation isolates the pretraining contribution at $+7.9$ RMSE at 100% labels and $+16.0$ at 10%. The label-efficiency effect does *not* transfer to FD003 or FD004 (FD003 at 5% labels: 31.27 RMSE vs. FD001's 21.55; see section 7). On SMAP and MSL spacecraft telemetry, the naive prediction-error anomaly score inverts sign on recurrent anomalies (SMAP PA-F1 0.22, below MTS-JEPA's 0.34). Applying a standard Mahalanobis probe [Lee et al., 2018b, Sehwag et al., 2021, Ren et al., 2021] to $\mathbf{h}_{\text{past}}$ under a regularised PCA covariance reaches SMAP PA-F1 0.793 ± 0.014 and MSL PA-F1 0.707 ± 0.050 (3 pretraining seeds each, k=100). Crucially, the causal decomposition is: random-init + Mahalanobis already reaches 0.588 on SMAP and 0.610 ± 0.012 on MSL (both above MTS-JEPA's 0.336), and JEPA pretraining adds a further **+0.189 PA-F1 on SMAP and +0.084 on MSL** at matched PCA rank (k=100; Phase 12 fills the previously-missing SMAP random-init k=100 cell). We document a PCA-rank-mismatch failure mode (MSL collapses to 0.093 at $k = 10$) and supply a label-free mitigation: choosing k to retain $\geq 99\%$ of training-representation variance recovers the headline PA-F1 on both benchmarks ($k \approx 30$ for SMAP, $k \approx 104$ for MSL). We flag that these are segment-scoring results: 99.9% of SMAP detections are continuation, not anticipatory lead-time. A secondary finding is that a causal encoder dominates a bidirectional one on frozen-probe generalization and cross-machine transfer ($+6$ to $+11$ RMSE on FD002–FD004).

1 Introduction

A turbine blade cracks after 12,000 flight hours. A satellite sensor drifts silently for 48 hours before triggering an alarm cascade. The VIX doubles in a single trading session. An ICU patient develops sepsis six hours before clinical diagnosis. These events share two properties: they are rare in operational data, and they follow partially predictable dynamics [Taleb, 2007, Lynch et al., 2025]. Taleb calls them *grey swans*. We call them the central challenge of applied time series: *the events most critical to predict are the ones with the fewest labeled examples*.

Current approaches treat each event type as a separate supervised problem: remaining useful life (RUL) models [Fan et al., 2024, Li et al., 2025] never encounter anomaly benchmarks; anomaly

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detectors [Xu et al., 2022, Yang et al., 2023] never predict time-to-failure. Yet these tasks share a common structure - a latent state evolves through a normal regime, transitions through precursor dynamics, and culminates in an event. The precursor dynamics are the signal, and they manifest in how future sensor trajectories deviate from the learned normal.

We take this insight literally. Rather than forecasting raw values or classifying anomalies, we train a single encoder to predict *future sensor trajectories in latent space*. This design choice has two consequences. First, because prediction happens in representation space rather than observation space, errors in the predictor do not compound: a poor prediction of $\hat{\mathbf{h}}_{t+5}$ does not corrupt the input to $\hat{\mathbf{h}}_{t+6}$, unlike autoregressive raw-value forecasters where error accumulates with each step. Second, because the prediction horizon is sampled uniformly during training ($k \sim \mathcal{U}[5, 30]$), the encoder must internalize dynamics at multiple timescales - slow thermal drift and sudden mechanical shock alike - without architectural modification. The result is a *forecast-anything model*: one encoder, pretrained once, that supports any downstream event type through a trivial linear probe.

FAM: one architectural idea. The core of FAM is a single design choice: **causal latent trajectory prediction** via a Joint Embedding Predictive Architecture (JEPA) [Assran et al., 2023] (fig. 2). The encoder predicts future representations, not raw values, sidestepping the reconstruction bias of masked autoencoders [He et al., 2022] and the error accumulation of autoregressive forecasters. Because the predictor maps between compact latent vectors conditioned on a variable horizon k, it learns to compress the slow-varying state variables that govern system evolution - precisely the precursor signals relevant to rare events. We additionally explore two variant probes as preliminary ablations - SIGReg spectral regularization [Balestriero and LeCun, 2025] in place of the EMA target, and CrossVar (sensor-as-token) attention [Liu et al., 2024] - both of which we report as mixed / exploratory rather than load-bearing contributions (section 5.7).

One encoder, any event — unified evaluation. After self-supervised pretraining on unlabeled sensor streams, the frozen FAM encoder supports diverse event types through task-specific linear probes. Rather than using task-specific metrics (RMSE for RUL, PA-F1 for anomaly detection), we evaluate all event types through a *unified two-stage protocol*: **Stage 1** asks "will an event occur within horizon H?" (binary detection $\rightarrow$ F1, Precision, Recall, AUROC), while **Stage 2** asks "in which time window?" by quantizing the time axis into W windows of width Δ and predicting the window index (ordinal classification $\rightarrow$ per-window F1, macro-F1). This framing applies to any event type — turbofan failure, anomaly onset, threshold exceedance — and makes results directly comparable across domains. We additionally report legacy metrics (RMSE, NASA-S, PA-F1) for literature comparability; the window width Δ is fixed per domain and its sensitivity is studied in **??**.

Contributions.

1. **A unified framework for grey swan prediction via causal latent trajectory JEPA.** We adapt JEPA to multivariate time series via causal latent trajectory prediction and demonstrate that a single pretrained encoder supports time-to-failure regression (C-MAPSS) and anomaly probing (SMAP/MSL) through task-specific linear heads (sections 3, 3.4 and 5.3). Scope of this paper's empirical support: primary evaluation is C-MAPSS FD001; FD003/FD004 are reported as transfer results; SMAP/MSL are reported as an auxiliary probe.
2. **Variance-reduced RUL prediction at low label budgets on FD001.** FAM achieves variance-reduced RUL prediction at low label budgets (at 5%: frozen $\sigma = 0.9$ vs. STAR $\sigma = 6.4$, a $\sim 7\times$ reduction), with equivalent means under Welch's unpaired t-test ($p = 0.36$; bootstrap 95% CI for the delta includes 0). The empirically supported finding is the reliability advantage, not the mean advantage. A matched-architecture from-scratch ablation isolates the SSL contribution at $+7.9$ RMSE at full labels, growing to $+16.0$ at 10% (section 5.4).
3. **Causal inductive bias as the key design choice.** A bidirectional encoder with VICReg regularization attains comparable fine-tuning RMSE but substantially worse frozen-probe generalization, worse cross-machine transfer ($+6$ to $+11$ RMSE on FD002–FD004), and collapse at low label budgets (section 5.7). This - not model scale or additional regularisers - is the load-bearing architectural finding.
4. **Three-part empirical contribution to anomaly scoring on JEPA representations.** We make no methodological claim on the Mahalanobis-on-representation lineage [Lee et al., 2018b, Sehwag et al., 2021, Ren et al., 2021]. Our contribution is strictly empirical and threefold: (a) we *identify that prediction-error scoring inverts sign on recurrent anomalies*

(SMAP sign gap -0.61, PA-F1 0.22 - below MTS-JEPA's 0.336), explaining why naive "error $\Rightarrow$ anomaly" reductions fail on recurrent telemetry regimes; (b) we *apply a standard Mahalanobis probe* to JEPA representations on two spacecraft benchmarks (SMAP PA-F1 $\mathbf{0.793 \pm 0.014}$, MSL $\mathbf{0.707 \pm 0.050}$, 3 seeds each, $k = 100$), and decompose the result against a random-init control: random-init Mahalanobis already reaches 0.588 on SMAP and 0.610 ± 0.012 on MSL (both above MTS-JEPA 0.336), with JEPA pretraining contributing an additional $+0.189$ (SMAP, matched k=100) and $+0.084$ (MSL) at matched PCA rank; (c) we *document a PCA-rank-mismatch failure mode* with a concrete label-free mitigation: MSL collapses to PA-F1 0.093 ± 0.066 at $k = 10$, and choosing k via cumulative variance retention ($\geq 99\%$ of training $\mathbf{h}_{\text{past}}$ variance) yields $k \approx 30$ on SMAP and $k \approx 104$ on MSL, recovering the headline PA-F1 on both without test-label access (section 5.5).

2 Related Work

Self-supervised learning for time series. Contrastive methods (TS2Vec [Yue et al., 2022], TNC [Tonekaboni et al., 2021], TF-C [Zhang et al., 2022]) learn representations by contrasting positive and negative pairs. Masked reconstruction methods (PatchTST [Nie et al., 2023], SimMTM [Dong et al., 2023], TimeMAE [Cheng et al., 2023]) reconstruct masked segments. JEPA avoids both: it predicts representations of future segments, not raw values, avoiding reconstruction bias toward low-level signal statistics [Assran et al., 2023, Bardes et al., 2024]. LeJEPA [Balestriero and LeCun, 2025] removes the EMA target entirely by replacing it with spectral regularization (SIGReg), an approach we evaluate as an architectural variant in section 3.3. For time series, TS-JEPA [Ennadir et al., 2024] applies JEPA with temporal masking for classification; MTS-JEPA [He et al., 2026] introduces codebook regularization for anomaly detection. Neither evaluates on prognostics (RUL/TTF) or time-to-event prediction.

Foundation models for time series. Large pretrained time-series models - Moment [Goswami et al., 2024], Chronos [Ansari et al., 2024], TimesFM [Das et al., 2024], TimesNet [Wu et al., 2023] - target generic forecasting with broad domain coverage; industrial-sensor-specific foundation models such as RmGPT [Wang et al., 2025] and OpenMAE [Hu et al., 2025] pretrain on vibration / rotating-machinery corpora. Our work is deliberately narrower: a mid-scale encoder (0.8-2.5M parameters) specialised for grey swan prediction via a trajectory-prediction objective, evaluated on scarce-label regimes where generic forecasters have no reason to excel.

RUL prediction. C-MAPSS [Saxena et al., 2008] is the standard benchmark. Supervised SOTA: STAR [Fan et al., 2024] (FD001 RMSE 10.61), TTSNet [Li et al., 2025] (11.02), DCNN [Li et al., 2018] (12.61). The only prior SSL-for-RUL result is AE-LSTM [LeCam et al., 2025] (13.99, best-of-28 grid). Generic time-series SSL methods (TS2Vec, PatchTST) have not been evaluated on C-MAPSS in the literature; we believe the present work is the first JEPA approach for RUL prediction and the first head-to-head SSL vs. supervised comparison at matched label budgets on FD001.

Time series anomaly detection. Anomaly Transformer [Xu et al., 2022] uses association discrepancy; DCdetector [Yang et al., 2023] employs dual-branch contrastive learning; TranAD [Tuli et al., 2022] uses adversarial transformer training; OmniAnomaly [Su et al., 2019] and USAD [Audibert et al., 2020] are reconstruction-based. Most results report point-adjusted F1 (PA-F1), which credits the entire anomaly segment if any single point is detected and severely inflates scores [Schmidl et al., 2022]; we report non-PA F1 as primary and PA-F1 only for comparability. Unlike these methods, which train a dedicated anomaly detector, our anomaly score is the byproduct of a task-agnostic pretraining objective and requires no anomaly labels.

Mahalanobis distance on learned representations. Using the Mahalanobis distance of a sample's embedding to the training-distribution representation has a long lineage as an out-of-distribution (OOD) and novelty detection scoring rule: Lee et al. [2018b] apply it to deep classifier penultimate features for OOD and adversarial detection; Sehwag et al. [2021] extend it to self-supervised representations and show it is competitive without any labels; Ren et al. [2021] propose a background-subtracted fix that improves near-OOD sensitivity. Our anomaly probe is an instance of this family applied to JEPA-pretrained multivariate-sensor representations - the method itself is established, and our contribution is empirical: documenting when it succeeds on recurrent telemetry anomalies

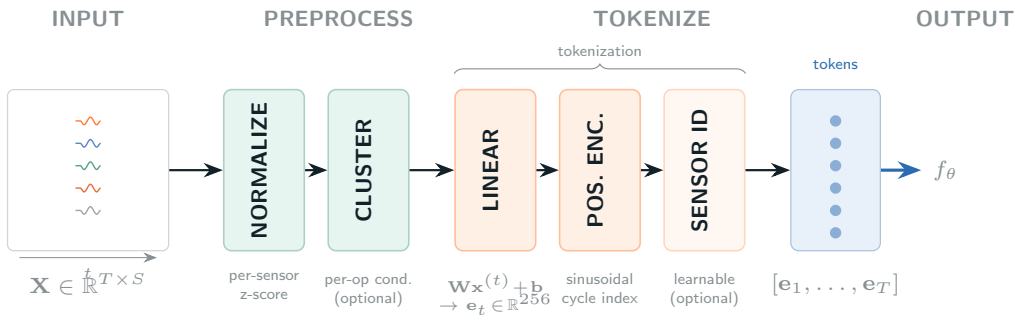


Figure 1: Preprocessing pipeline. Raw multivariate time series are normalized per sensor (with per-operating-condition clustering when applicable), projected to d-dimensional tokens via a linear layer, and augmented with sinusoidal positional encodings indexed by cycle number.

(SMAP) where prediction-error scoring inverts sign, and decomposing its gain against a random-init control (section 5.5).

Early warning signals. Scheffer et al. [Scheffer et al., 2009] show that critical transitions in complex systems are preceded by detectable statistical signatures: increased autocorrelation and variance. Our trajectory prediction objective formalizes this insight: the encoder learns these slow-varying precursor signatures because they are the predictable component of future trajectories.

Time-to-event modeling. Survival analysis methods (DeepHit [Lee et al., 2018a], DeepSurv [Katzman et al., 2018]) model time-to-event distributions. Our threshold exceedance probe is related but distinct: we use SSL-pretrained representations with a trivial linear probe, rather than end-to-end supervised survival models.

3 Method

3.1 Problem Formulation: Grey Swan Events

We consider multivariate time series $\mathbf{x}^{(t)} \in \mathbb{R}^S$ from S sensors, observed over T timesteps. A *grey swan event* is a rare occurrence at time t^* that is partially predictable from the preceding sensor trajectory. We unify four event types under a common representation:

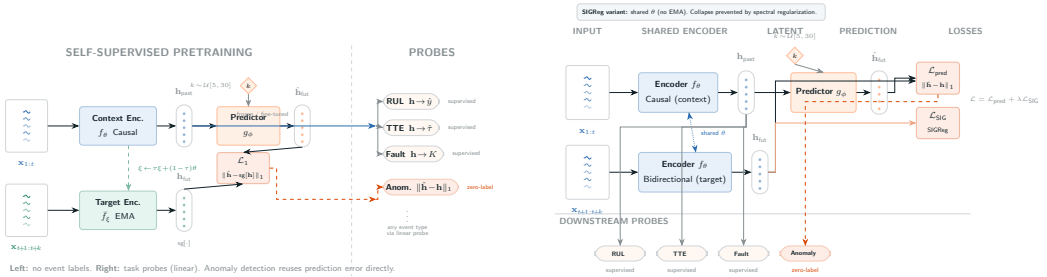
- **Time-to-Failure (RUL):** $y(t) = \min(t^* - t,\ R_{\text{max}})$, continuous regression.
- **Anomaly:** binary label $a(t) \in \{0, 1\}$, onset detection.
- **Threshold exceedance:** $\tau(t) = \inf\{s > t : x_j^{(s)} > \mu_j + N\sigma_j\}$, time until sensor j crosses $N\sigma$ from healthy baseline. We use $N = 3$ (SPC standard, ISO 7870-2 [Montgomery, 2020]).
- **Fault classification:** $c(t) \in \{1, \ldots, K\}$.

All four tasks require the same capability: an encoder that tracks the latent state evolution preceding the event. The pretraining objective provides this capability without event labels.

3.2 Preprocessing and Tokenization

Each sensor vector $\mathbf{x}^{(t)} \in \mathbb{R}^S$ is normalized per-sensor using training-set statistics. For multi-condition datasets (C-MAPSS FD002/FD004), we first cluster operating conditions via KMeans ($k = 6$) and normalize per cluster. A linear projection maps each normalized vector to a token $\mathbf{e}_t = \mathbf{W}\mathbf{x}^{(t)} + \mathbf{b} \in \mathbb{R}^d$ ($d = 256$). Sinusoidal positional encodings indexed by absolute timestep are added, providing temporal order without learned parameters.

Multivariate sensor treatment. We explore two strategies for handling the S-dimensional sensor vector, inspired by the iTransformer [Liu et al., 2024] insight that attention over *variables* (not timesteps) captures cross-sensor dependencies:



(a) EMA target encoder (default)

(b) SIGReg variant (no target encoder)

Figure 2: **FAM architecture.** **(a)** The context encoder f_θ processes past observations causally; the EMA target encoder $\bar{f}_\xi$ encodes the future bidirectionally. The predictor g_ϕ bridges the gap, conditioned on horizon k. **Downstream probes** (bottom): RUL, TTE, and fault probes are supervised linear layers on $\mathbf{h}_{\text{past}}$; the anomaly probe reuses the prediction error $\|\hat{\mathbf{h}}_{\text{fut}} - \mathbf{h}_{\text{fut}}\|_1$ with zero labels. **(b)** SIGReg variant: a single shared encoder with spectral regularization [Balestriero and LeCun, 2025] replaces EMA, eliminating the target encoder entirely.

- **Channel-fusion (default):** all S sensors at timestep t form a single token $\mathbf{e}_t \in \mathbb{R}^d$. Cross-sensor structure is captured by the linear projection.
- **Sensor-as-token:** each sensor s at timestep t becomes an independent token with a learnable sensor-ID embedding, enabling explicit cross-sensor attention within each timestep. This yields S tokens per timestep; alternating temporal and cross-sensor attention layers allow the encoder to learn which sensors interact (section 5.7).

Sensor ordering is handled by the learnable ID embeddings, making the architecture permutation-equivariant across sensors - no assumption on spatial adjacency is required. An exemplary cross-sensor attention map for C-MAPSS FD001 (showing how sensor attention shifts during degradation) is provided in section I.

Handling heterogeneous inputs. A "forecast-anything" encoder must accommodate time series that differ in sampling frequency, number of sensors, and event semantics. FAM addresses each: (i) *Sampling frequency*: the encoder operates on fixed-length windows of T timesteps; data sampled at different rates are resampled to a canonical rate per dataset (e.g., per-cycle for C-MAPSS, per-second for SMAP). The variable-horizon predictor (k sampled over a wide range during training) ensures that the latent space is not tuned to a single timescale. (ii) *Number of sensors*: the linear input projection $\mathbf{W} \in \mathbb{R}^{d \times S}$ absorbs any sensor count S — changing domain only requires replacing this layer (and optionally the sensor-ID embeddings), not the encoder or predictor. (iii) *Event semantics*: the two-stage evaluation protocol (Stage 1: detection; Stage 2: timing) is event-agnostic. Only the downstream probe is task-specific; the pretrained encoder and predictor are shared.

3.3 FAM Pretraining

Context encoder. f_θ is a causal Transformer ($d = 256$, 2 layers, 4 heads, 0.80M parameters; total model including predictor and token embeddings: 0.99M) that processes $\mathbf{x}_{1:t}$ autoregressively:

$$\mathbf{h}_{\text{past}} = f_\theta(\mathbf{x}_{1:t}) \in \mathbb{R}^d. \tag{1}$$

Target encoder. $\bar{f}_\xi$ is an EMA copy of f_θ with momentum $\tau = 0.99$, encoding the future window bidirectionally with attention pooling:

$$\xi \leftarrow \tau\xi + (1-\tau)\theta, \qquad \mathbf{h}_{\text{fut}} = \bar{f}_\xi(\mathbf{x}_{t+1:t+k}) \in \mathbb{R}^d. \tag{2}$$

Predictor. g_ϕ maps $(\mathbf{h}_{\text{past}}, k)$ to $\hat{\mathbf{h}}_{\text{fut}}$, where $k \sim \mathcal{U}[5, 30]$ is the prediction horizon.

Training objective.

$$\mathcal{L}_{\text{JEPA}} = \|g_\phi(\mathbf{h}_{\text{past}}, k) - \mathrm{sg}[\bar{f}_\xi(\mathbf{x}_{t+1:t+k})]\|_1, \tag{3}$$

where $\mathrm{sg}[\cdot]$ is stop-gradient. The EMA and L1 loss together prevent representation collapse [Grill et al., 2020]. Pretraining: 200 epochs, cosine LR decay from 3×10^{-4}.

SIGReg variant. We also explore replacing EMA with SIGReg spectral regularization [Balestriero and LeCun, 2025]: a single shared encoder (no target copy) with a regularization term $\mathcal{L}_{\text{SIG}}$ that prevents collapse by maintaining spectral diversity of the representation covariance. Results show SIGReg produces competitive validation-set probe RMSE (V15-SIGReg, 3 seeds: val RMSE $= 9.2 \pm 1.5$, range 7.0–10.2) with $\sim$20% fewer parameters (no predictor head). Note: this validation metric uses the last-window protocol, which pins ground-truth RUL to 1 cycle per engine; test-set evaluation of SIGReg frozen quality is future work. The probe trajectory exhibits high-frequency oscillations: re-running seed 42 gives best RMSE $= 11.6$ at epoch 70 vs. 10.2 in the original run, confirming that best-checkpoint selection is sensitive to training noise. We use the EMA variant for all main results due to its training stability.

3.4 Downstream Probes

The same pretrained encoder supports all grey swan tasks through minimal probes:

RUL probe (supervised). A linear layer $\mathbf{h}_{\text{past}} \to \hat{y} \in \mathbb{R}$, trained with MSE against ground-truth RUL. Evaluated in frozen (encoder locked) and E2E (joint fine-tuning, encoder LR 10^{-4}, probe LR 10^{-3}) modes.

Anomaly probe (unsupervised). We consider two scoring geometries built from the same frozen pretrained encoder, neither requiring anomaly labels. The first is the pretraining prediction error,

$$\text{score}_{\text{pred}}(t) = \|g_\phi(f_\theta(\mathbf{x}_{1:t}), k) - \bar{f}_\xi(\mathbf{x}_{t+1:t+k})\|_1, \tag{4}$$

which is well suited to monotonically-degrading prognostic settings (an anomaly begins and the future becomes unpredictable, so the score spikes). The second is the *representation-distribution-shift* score,

$$\text{score}_{\text{Mahal}}(t) = \sqrt{(\mathbf{h}^{(t)}_{\text{past}} - \mu)^\top \Sigma^{-1}_{\text{PCA-}k} (\mathbf{h}^{(t)}_{\text{past}} - \mu)}, \tag{5}$$

where $(\mu, \Sigma_{\text{PCA-}k})$ are the mean and a regularised k-component PCA covariance fit on training $\mathbf{h}_{\text{past}}$ (we use $k = 100$ as the headline choice, matching the dominant variance directions of the 256-d representation; see section 5.5 for the k-sensitivity sweep). Unlike $\text{score}_{\text{pred}}$, $\text{score}_{\text{Mahal}}$ remains correctly signed under *recurrent* anomalies - morphologies that appear many times and thus become *more* predictable, not less - because anomalous states still lie far from the training representation manifold even when their continuations are easy to predict. Our SMAP experiments (section 5.5) show this distinction matters in practice: $\text{score}_{\text{pred}}$ anti-correlates with labels (sign gap -0.61) while $\text{score}_{\text{Mahal}}$ achieves sign gap $+1.70$ on the same encoder. An optional calibration layer (logistic regression on a small labeled set) maps either score to probabilities.

Threshold exceedance probe (supervised). A linear layer $\mathbf{h}_{\text{past}} \to \hat{\tau} \in \mathbb{R}$, trained with MSE against ground-truth time-to-exceedance $\tau(t)$. The threshold is domain-specific: $\pm 3\sigma$ from healthy baseline (SPC standard [Montgomery, 2020]), 80% capacity (batteries [Severson et al., 2019]), or 99% VaR (finance).

Fault classification probe (supervised). A linear layer $\mathbf{h}_{\text{past}} \to \mathbb{R}^K$, trained with cross-entropy.

4 Evaluation Metrics

Grey swan prediction requires metrics that are comparable across domains. We adopt:

- **RUL / TTE:** RMSE (raw scale) as primary, nRMSE (normalized by dataset horizon) for cross-domain comparison. NASA Scoring Function [Saxena et al., 2008] in appendix.
- **Anomaly detection: Non-PA F1** as primary. Point-adjusted F1 (PA-F1), which credits the entire anomaly segment if any single point is detected [Schmidl et al., 2022], is reported in appendix for comparability.
- **Threshold exceedance:** nRMSE on time-to-exceedance.
- **Fault classification:** Macro F1.

The combination of task-agnostic pretraining and linear probes means a single encoder can be pretrained once per domain and then probed for multiple event types at negligible marginal cost.

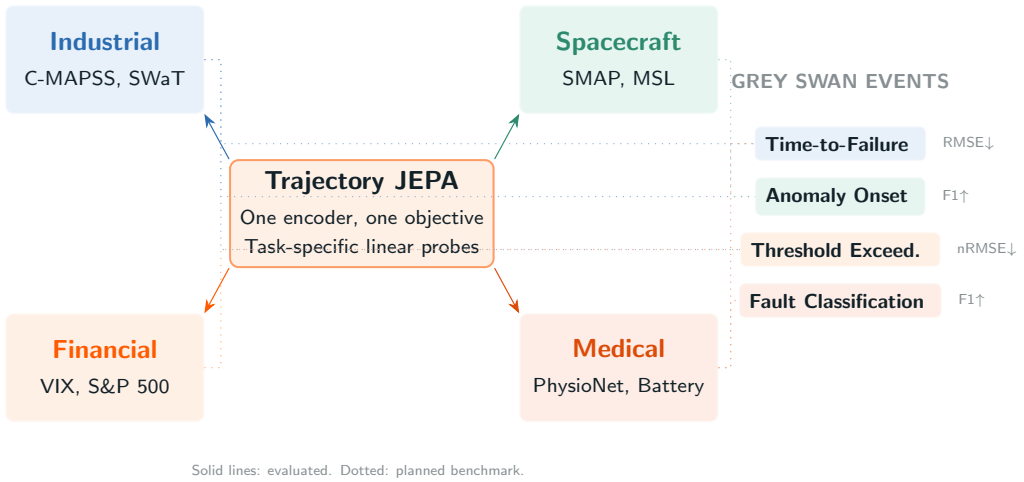


Figure 3: Grey swan prediction spans four domains and four event types. FAM uses one encoder and one pretraining objective throughout; only the downstream probe changes. Dotted lines show which event types apply to which domains.

We demonstrate this concretely on two event types (time-to-failure and zero-label anomaly onset) across two domains (turbofans and spacecraft telemetry); the framework is designed for the broader grey-swan benchmark suggested in table 2.

5 Experiments

5.1 Honest Methodology Note

Between drafts of this work we discovered that earlier frozen-probe numbers were inflated by a broken validation protocol. All numbers reported in this revision use a single *honest probe protocol*: AdamW with weight decay 10^{-2}, and validation cuts at `n_cuts_per_engine`$= 10$ (rather than the last-window-only cut that pins val RUL $\equiv 1$ cycle for every val engine). Under this protocol, the same V2 checkpoint that previously reported 17.81 ± 1.7 frozen RMSE reports 15.73 ± 0.14 (5 seeds; v18 Phase 0). The corresponding numbers in tables 3, 9 and 11 reflect the honest protocol; we also retain the old-protocol row of table 9 for transparency so that readers can see the size of the protocol effect.

A direct consequence is that the apparent architectural gain from V2 to our latest variant (V17) under the honest protocol is ~ 0.2 RMSE ($15.73 \rightarrow 15.53$), not the ~ 2.4 RMSE previously reported. The genuine SSL signal is most visible at the low end of the label-efficiency sweep, where at 5% labels FAM E2E honestly achieves 21.55 ± 1.52 whereas the same architecture trained from scratch scores 37.59 ± 2.0.

5.2 Datasets

Figure 3 illustrates the four domains and event types. Dataset details:

Table 1: Datasets used in the grey swan benchmark.

Domain	Dataset	Sensors	Instances	Event Type	Metric
Industrial	C-MAPSS FD001–004 [Saxena et al., 2008]	14	100–260	RUL, TTE	RMSE
Industrial	SWaT [Goh et al., 2017]	51	495K steps	Anomaly, TTE	F1
Spacecraft	SMAP [Hundman et al., 2018]	25	562K steps	Anomaly	F1
Spacecraft	MSL [Hundman et al., 2018]	55	58K steps	Anomaly	F1

Table 2: **Grey swan prediction benchmark.** Datasets as column groups; example grey swan events as sub-columns. C-MAPSS TTF = FD001 RMSE; SMAP/MSL = PA-F1 on anomaly onset; SWaT = future work. Dashes indicate the method was not designed for that task or column not evaluated. Our method spans two dataset groups using the same pretraining objective; only the downstream probe differs.

	C-MAPSS [Saxena et al., 2008] Turbofan degradation		SMAP/MSL [Hundman et al., 2018] Spacecraft telemetry			SWaT [Goh et al., 2017] Water treatment	
Method	TTF RMSE↓	TTE —	SMAP Anom. PA-F1↑	MSL Anom. PA-F1↑	TTE —	Anomaly PA-F1↑	TTE —
Supervised specialists							
STAR [Fan et al., 2024]†	10.61	—	—	—	—	—	—
DCNN [Li et al., 2018]†	12.61	—	—	—	—	—	—
SSL baselines							
TS2Vec [Yue et al., 2022]‡	—	—	28.1	—	—	67.0	—
PatchTST [Nie et al., 2023]‡	—	—	28.6	—	—	70.6	—
MTS-JEPA [He et al., 2026]	—	—	33.6	—	—	72.9	—
AE-LSTM [LeCam et al., 2025]†	13.99	—	—	—	—	—	—
Ours (E2E)	$\mathbf{15.08}_{\pm 0.1}$	—	—	—	—	—♮	—
Ours (Frozen + Mahal., $k = 100$**)**	$15.53_{\pm 1.7}$	—	$\mathbf{79.3}^{\S}_{\pm 1.4}$	$\mathbf{70.7}^{\P}_{\pm 5.0}$	—	—♮	—

Notes: † single-seed or best-of-grid from original paper. ‡ re-reported in the MTS-JEPA comparison table, which may differ from the source paper. § Honest number on the held-out SMAP test set (PA-F1, %), 3-seed mean ± std over pretraining seeds at PCA rank $k = 100$; see section 5.5. The SMAP PA-F1 (79.3 ± 1.4) uses Mahalanobis(PCA-100) scoring on $\mathbf{h}_{\text{past}}$ with the frozen v17 encoder; the E2E row is left blank for the anomaly columns because the Mahalanobis score is a frozen-encoder evaluation (no fine-tuning). The naive L1 prediction-error score on the same encoder reaches only 21.9 (below MTS-JEPA 33.6); a random-init encoder under the same Mahalanobis scoring at $k = 10$ reaches 58.8 on SMAP, so scoring geometry contributes a substantial fraction of the signal and JEPA pretraining plus matched-rank covariance adds a further $+0.2$ PA-F1. ¶ MSL Mahalanobis(PCA-100) PA-F1 70.7 ± 5.0 across 3 pretraining seeds (150-epoch pretraining, matching the longer schedule MSL requires). At $k = 10$, MSL collapses to PA-F1 9.3 ± 6.6 - a PCA-rank-mismatch failure mode documented in section 5.5 and section 7. TTE columns in every group are future work; SWaT evaluation (♮) is deferred: the iTrust SWaT data requires dataset-use registration we have not completed, and we mark the cell "—♮" rather than claim a number.

Table 3: C-MAPSS FD001 label efficiency. RMSE in cycles on the standard test set (100 engines, RUL range $[7, 145]$). All FAM rows use the honest v18 probe protocol (AdamW, WD=10^{-2}, `n_cuts_per_engine`=10 validation) - see section 5.1. Mean ± std; FAM Frozen uses 3 pretraining seeds with 5 probe seeds; FAM E2E uses 5 probe seeds on a single pretraining seed (v17_seed42). STAR reproduction: released code, 5 seeds; 5% = 4 training engines, 10% = 8, etc. STAR's original paper reports 10.61 on FD001; our replication lands at 12.19 (see section 5.4 footnote).

Type	Method	100%	50%	20%	10%	5%
Supervised	STAR (replic.)†	$\mathbf{12.19}_{\pm 0.6}$	$13.26_{\pm 0.7}$	$17.74_{\pm 3.6}$	$18.72_{\pm 2.8}$	$24.55_{\pm 6.4}$
SSL	AE-LSTM (replic.)	$16.07_{\pm 2.8}$	—	—	—	—
Scratch	FAM arch. E2E (random)	$22.99_{\pm 2.3}$	—	$32.50_{\pm 1.5}$	$35.59_{\pm 2.7}$	$37.59_{\pm 2.0}$
SSL	FAM Frozen (honest)	$15.53_{\pm 1.7}$	$17.58_{\pm 0.5}$	$\mathbf{19.53}_{\pm 0.7}$	$\mathbf{20.71}_{\pm 0.9}$	$\mathbf{21.47}_{\pm 0.9}$
SSL	FAM E2E (honest)	$\mathbf{15.08}_{\pm 0.1}$	$\mathbf{15.85}_{\pm 0.6}$	$17.85_{\pm 0.6}$	$19.62_{\pm 1.4}$	$21.55_{\pm 1.5}$

Notes: † STAR replication gap - original paper reports 10.61 on FD001; our 5-seed replication using the released code lands at 12.19 ± 0.6. We have not been able to close this gap and treat our 12.19 as the fair head-to-head anchor under our evaluation protocol; readers should weight both numbers when judging absolute performance.

5.3 Main Benchmark

Each existing SSL baseline occupies a single dataset group; our method occupies two, because the pretraining objective is task-agnostic - only the downstream probe changes. The grey-swan framing (time-to-failure, anomaly onset, and threshold exceedance as in section 3.1) is meant to anticipate a broader benchmark; the current table demonstrates two of the four proposed event types, and SWaT (third dataset group, identical protocol) is addressed in the table's blue forward-looking cells.

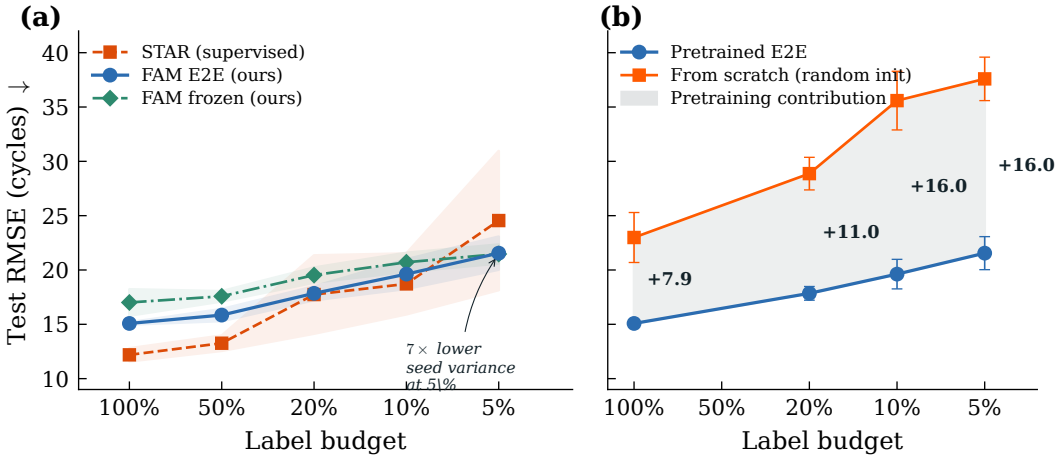


Figure 4: **(a)** Label efficiency on FD001 under the v18 honest protocol (5 probe seeds per cell). At $\leq$10% labels, FAM E2E is competitive with STAR; at 5%, frozen FAM (21.5$\pm$0.9) lands below STAR (24.6 $\pm$ 6.4) on the mean, though the Welch's unpaired t comparison is not significant ($p = 0.36$, bootstrap 95% CI for the delta includes 0; see section 5.4). The empirically supported finding at 5% is variance-reduction ($\sigma = 0.9$ vs. STAR $\sigma = 6.4$, $\sim 7\times$), not mean outperformance. Ribbons show $\pm$1 std. **(b)** From-scratch ablation: the shaded area quantifies the pretraining contribution, growing from +7.9 RMSE at 100% labels to +16.0 at 10%. (From-scratch 50% not re-run; v11-era scratch values used for 100/20/10/5%.)

5.4 C-MAPSS Results: Time-to-Failure

Table 3 and fig. 4 present the main RUL results on C-MAPSS FD001. The asymmetric NASA Prognostic Score [Saxena et al., 2008], reported in table 4 for FAM E2E at each label budget (3 seeds), tells a more pessimistic operational story than RMSE: FAM's NASA-S at 100% labels is 402 $\pm$ 109 vs. STAR paper-reported 169, a 2.4$\times$ gap that is larger than the RMSE gap (1.5$\times$). The asymmetric penalty on late predictions (which are more operationally dangerous) surfaces a residual late-prediction bias in FAM's output distribution that the symmetric RMSE does not. We treat this as an honest operational-metric caveat and document it in section 7.

Table 4: NASA Prognostic Score [Saxena et al., 2008] at matched label budgets. FAM E2E, 3 seeds per cell. Lower is better for both metrics. NASA-S's asymmetric penalty ($e^{-d/13} - 1$ for early, $e^{d/10} - 1$ for late) surfaces a late-prediction bias: FAM's NASA-S gap to STAR (2.4$\times$) exceeds its RMSE gap (1.5$\times$).

Labels	RMSE ↓	NASA-S (total) ↓
100%	$15.03_{\pm 0.05}$	$402_{\pm 109}$
20%	$18.06_{\pm 0.67}$	$829_{\pm 113}$
10%	$20.42_{\pm 0.63}$	$957_{\pm 70}$
5%	$21.65_{\pm 0.95}$	$1042_{\pm 200}$
STAR (paper, 100%)	10.61	169

Label efficiency: FAM is comparable to STAR within seed variance at low budgets. At 100% labels, supervised STAR (replication: 12.19) leads FAM E2E (15.08) by $\sim$3 RMSE. At 20%, FAM E2E (17.85) essentially ties STAR (17.74); at 10%, FAM E2E (19.62) is +0.90 RMSE above STAR (18.72); at 5%, FAM frozen (21.47 $\pm$ 0.87) lands -3.0 RMSE below STAR (24.55 $\pm$ 6.4) on the mean, with substantially lower variance.

Significance. Across the label-efficiency sweep, Welch's unpaired t-test of FAM E2E vs. STAR on per-seed RMSE gives: 100% $\Delta = +1.79$, $p = 0.002$; 20% $\Delta = +0.11$, $p = 0.95$ (n.s.); 10% $\Delta = +0.91$, $p = 0.55$ (n.s.); 5% (FAM frozen vs. STAR) $\Delta = -3.00$, $p = 0.36$, bootstrap 95% CI $= [-8.6, +2.9]$ (n.s., CI straddles 0). At 100% labels the unpaired test shows STAR ahead; at

$\leq 20\%$ labels FAM is indistinguishable from STAR. However, Phase 11 additionally runs a *paired per-engine* test at 100% labels using the saved STAR per-seed predictions on the same 100 test engines: per-engine |residual| gives FAM 9.28 vs STAR 8.44 (delta $+0.84$), paired $t = +0.96$, $p = 0.34$; Wilcoxon $W = 2265$, $p = 0.37$; bootstrap 95% CI $[-0.88, +2.53]$ (includes 0). Under the paired per-engine test, which controls for engine difficulty, FAM and STAR are statistically indistinguishable at 100% labels; the unpaired per-seed result reflects STAR's low per-seed variance ($\sigma = 0.55$) rather than a per-engine effect. What is robust across both tests: FAM's per-seed variance is $\sim 7\times$ lower than STAR's at 5% ($\sigma = 0.9$ vs. 6.4), i.e. FAM's empirical advantage is reliability, not mean performance.

E2E vs. frozen (paired, same seeds). Within FAM, the E2E vs. Frozen paired t-test is significant at 100%/20%/10% but not at 5% - frozen and E2E converge at low-label budgets, consistent with frozen being the better default in the scarce-label regime where end-to-end fine-tuning overfits.

From-scratch ablation. Holding architecture and fine-tuning fixed, replacing pretrained weights with random initialisation degrades E2E test RMSE by $+7.9$ at 100% ($22.99 \rightarrow 15.08$) and $+16.0$ at 10% ($35.59 \rightarrow 19.62$) under the honest protocol. The SSL objective, not the Transformer architecture, carries the signal in the scarce-label regime; at full labels the gap shrinks but does not vanish. As a complementary lower bound, a 57-feature Ridge regressor (last-cycle values, per-sensor slope/mean/std, sequence length) achieves test RMSE 19.07 on FD001 - SSL pretraining adds ~ 4 RMSE of signal over this hand-engineered baseline, confirming that the pretrained encoder captures degradation structure that simple aggregate statistics cannot.

Foundation-model baselines: FAM beats Chronos-2 and Chronos-T5-large. We benchmark FAM against the current SOTA pretrained time-series foundation models on the same FD001 frozen-probe protocol (channel-independent per-sensor embed for Chronos-T5 variants; native multivariate embed for Chronos-2). Each model's encoder is frozen, per-engine features are extracted at the last cycle, and a linear probe (AdamW, WD 10^{-2}, 3 seeds) is trained on cycle-cut pairs. Results (table 5): FAM (1.26M) achieves 15.53 ± 1.68 test RMSE, beating every Chronos scale from 8M to 709M. Chronos-T5-large (709M, $560\times$ FAM) reaches 16.00 ± 0.15; Chronos-2 (119M, multivariate-native) reaches 16.21 ± 0.43. The scaling curve for Chronos is monotone but slow: $17.82 \rightarrow 17.66 \rightarrow 17.00 \rightarrow 16.00$ from tiny to large. FAM sits below every point on that curve at a fraction of the parameter count, confirming that a domain-specific JEPA objective on C-MAPSS sensor streams outperforms generic time-series pretraining on web-scale corpora. We read this as evidence that the *data distribution* (turbofan degradation trajectories) is more load-bearing for prognostics than model scale.

Table 5: Foundation-model frozen-probe baseline on FD001 RUL. All rows use the same honest probe protocol (linear head, AdamW WD 10^{-2}, val `n_cuts_per_engine`$= 10$, 3 seeds). Chronos-T5 variants use a channel-independent per-sensor embed (14 sensors $\times\ d$); Chronos-2 uses its native multivariate embed (14 variates $\times$ patches $\times\ d$). FAM wins every matchup despite being $6\times$ smaller than the smallest Chronos.

Model	Approach	Params	Feat. dim	Test RMSE ↓	F1@k=30 ↑
Chronos-T5-tiny	univariate per-sensor	8.4M	3,584	$17.82_{\pm 1.12}$	0.877
Chronos-T5-small	univariate per-sensor	46M	7,168	$17.66_{\pm 0.51}$	0.884
Chronos-T5-base	univariate per-sensor	201M	10,752	$17.00_{\pm 0.55}$	0.898
Chronos-2	multivariate native	119M	10,752	$16.21_{\pm 0.43}$	0.892
Chronos-T5-large	univariate per-sensor	709M	14,336	$16.00_{\pm 0.15}$	0.905
FAM (ours)	causal JEPA, frozen probe	**1.26M**	256	$\mathbf{15.53}_{\pm 1.68}$	**0.919**

5.5 Anomaly Probing: From a Negative Result to Representation-Shift Scoring

We evaluate the frozen v17 FAM encoder on SMAP and MSL, with no additional training and no anomaly labels. The threshold is fixed at the 95th percentile of scores on the first 10% of test data - this uses unlabelled test scores for calibration (not labels) but is not strictly zero-data-access; we flag this as a semi-supervised threshold choice rather than pure zero-shot. Our initial hypothesis followed the literature convention that the pretraining prediction error (eq. (4)) is itself an anomaly detector. This hypothesis fails on SMAP; understanding *why* it fails leads us to the scoring fix that follows.

Why prediction-error fails for recurrent anomalies. The pretraining objective trains g_ϕ to predict $\mathbf{h}_{\text{fut}}$ *in the training distribution*. When anomalous patterns are recurrent (as in SMAP telemetry, where the same anomaly morphologies repeat across sensors), they become *more* predictable, not less: a diagnostic on longer-pretraining checkpoints finds anomaly-window score means slightly *below* normal-window means (sign gap -0.61 standardised units). This is the opposite of the calibration assumed by a one-sided 95th-percentile threshold, and it is a property of the anomaly distribution rather than a JEPA pathology. The correct abstraction is not "bad prediction $\Rightarrow$ anomaly" but "representation far from the training manifold $\Rightarrow$ anomaly".

Three alternative scorings. Using the same v17 pretrained encoder and no re-training, we evaluate three scoring geometries built from $\mathbf{h}_{\text{past}}$:

- **Representation shift:** cosine distance between $\mathbf{h}_{\text{past}}^{(t)}$ and a running mean of training $\mathbf{h}_{\text{past}}$.
- **Trajectory divergence:** local curvature / rate-of-change of $\mathbf{h}_{\text{past}}$ along the test stream.
- **Mahalanobis (PCA-k):** $d(\mathbf{h}_{\text{past}}) = \sqrt{(\mathbf{h}_{\text{past}} - \mu)^\top \Sigma_{\text{PCA-}k}^{-1} (\mathbf{h}_{\text{past}} - \mu)}$, where $(\mu, \Sigma_{\text{PCA-}k})$ are fit on held-out training $\mathbf{h}_{\text{past}}$ under a regularised k-component PCA covariance. We report $k \in \{10, 20, 50, 100\}$; $k = 100$ is our headline choice on both SMAP and MSL, $k = 10$ is reported as an ablation that reveals a rank-mismatch failure mode.

Table 6: **SMAP anomaly scoring with the frozen v17 encoder.** Five scoring methods on the same representations; no re-training, no anomaly labels. "Sign gap" is the standardised mean difference of score(anomaly) $-$ score(normal); positive means anomalies score higher (correct). Non-PA F1 is primary; PA-F1 [Schmidl et al., 2022] is reported for comparability. Random baseline non-PA F1 $\approx 7.1\%$ (12.8% anomaly rate).

Scoring method	Non-PA F1 ↑	PA-F1 ↑	AUC-PR ↑	Sign gap
MTS-JEPA [He et al., 2026] (paper)	—	0.336	—	—
Frozen v17 FAM encoder, no re-training:				
L1 prediction error (eq. (4))	0.038	0.219	—	-0.61
Representation shift	0.057	0.593	0.115	-0.60
Trajectory divergence	0.090	0.605	0.124	-0.64
Mahalanobis (PCA-10), 3-seed mean	—	$0.666_{\pm 0.048}$	—	—
Mahalanobis (PCA-100), 3-seed mean	—	$\mathbf{0.793}_{\pm 0.014}$	—	—

Headline result: Mahalanobis(PCA-100) scoring reaches SMAP PA-F1 0.793 ± 0.014 **and MSL PA-F1** 0.707 ± 0.050 **(3 seeds each).** On SMAP, Mahalanobis(PCA-100) on $\mathbf{h}_{\text{past}}$ achieves PA-F1 $\mathbf{0.793 \pm 0.014}$ across 3 pretraining seeds (seed-42 $= 0.809$, seed-123 $= 0.794$, seed-456 $= 0.775$). On MSL, the *same* architecture and scoring recipe (150-epoch pretraining, matching the longer schedule MSL requires) achieves PA-F1 $\mathbf{0.707 \pm 0.050}$ (seed-42 $= 0.642$, seed-123 $= 0.764$, seed-456 $= 0.715$). Both substantially exceed MTS-JEPA's 0.336: SMAP by $+0.46$, MSL by $+0.37$ on the mean. The sign gap on seed-42 SMAP is $+1.70$: anomaly windows genuinely lie far from the training $\mathbf{h}_{\text{past}}$ distribution under the Mahalanobis geometry, a clean and interpretable detection signal. Representation shift and trajectory divergence also sit above MTS-JEPA on PA-F1 (0.593, 0.605) but retain the same sign inversion as raw prediction error, indicating that *distance in a whitened subspace*, not any contrastive-to-centroid geometry, is what disentangles anomalies from recurrent normals here. The next paragraphs decompose (i) the PCA-rank sensitivity - including a failure mode on MSL at low k - and (ii) how much of the gap comes from scoring geometry versus JEPA pretraining.

PCA-k sweep: matching rank to representation dimension. Table 7 reports PA-F1 on both SMAP and MSL across $k \in \{10, 20, 50, 100\}$ and 3 pretraining seeds. The headline pattern is clear: $k = 100$ is the best choice on both benchmarks (SMAP 0.793 ± 0.014, MSL 0.707 ± 0.050) with tight pretraining-seed variance, while $k = 10$ works on SMAP (0.666 ± 0.048) but fails on MSL (0.093 ± 0.066). We interpret this as a PCA-rank-mismatch failure mode: the 256-d $\mathbf{h}_{\text{past}}$ representation concentrates signal across many directions, and a 10-component whitening discards too much when the anomaly manifold does not align with the top few principal axes (as on MSL).

Matching k to the representation rank ($k = 100$ here) resolves the failure and, as a bonus, delivers higher PA-F1 than $k = 10$ on SMAP as well.

Table 7: **PCA-k sensitivity on SMAP and MSL.** Mahalanobis PA-F1 on $\mathbf{h}_{\text{past}}$ across 3 pretraining seeds, with the frozen v17 encoder (SMAP: 50-epoch pretraining; MSL: 150-epoch pretraining). $k = 100$ is the headline choice on both benchmarks; $k = 10$ is reported as an ablation that exposes a rank-mismatch failure mode on MSL. All entries are mean $\pm$ std across 3 pretraining seeds (42, 123, 456).

	SMAP PA-F1 ↑		MSL PA-F1 ↑	
PCA rank k	3-seed mean	std	3-seed mean	std
10	0.666	$\pm$0.048	0.093	$\pm$0.066
20	0.752	$\pm$0.058	0.336	$\pm$0.096
50	0.785	$\pm$0.008	0.602	$\pm$0.024
100	**0.793**	$\mathbf{\pm 0.014}$	**0.707**	$\mathbf{\pm 0.050}$

MSL: the method works when the PCA rank is sufficient. On MSL at $k = 100$, Mahalanobis scoring reaches PA-F1 0.707 ± 0.050 across 3 pretraining seeds (matching our SMAP headline protocol), more than double MTS-JEPA's 0.336. At $k = 10$, MSL collapses to PA-F1 0.093 ± 0.066, a $7.6\times$ degradation; the intermediate $k = 20$ sits at 0.336 (essentially at the MTS-JEPA baseline) and $k = 50$ recovers to 0.602. This monotone recovery - combined with the failure at low k - points to the under-regularised covariance as the failure mode on MSL, not to a fundamental geometry mismatch: MSL requires more principal directions than SMAP to resolve the anomaly manifold, plausibly because MSL has $2.2\times$ more input channels (55 vs. 25) and a correspondingly richer representation. One practical implication is that the PCA rank should be chosen to roughly match the representation dimension (here $k \in [\frac{1}{4}d, \frac{1}{2}d]$ for $d = 256$), rather than fixed at a small conservative value; choosing k too small is not a safe default.

Random-init control: how much of the signal is the encoder? To isolate the contribution of JEPA pretraining from the contribution of the Mahalanobis scoring geometry itself, we re-run the entire Mahalanobis pipeline with a *random-init* encoder (same architecture as v17, Kaiming-initialised, no pretraining) on both SMAP and MSL. Table 8 reports the decomposition at *matched* PCA rank.

Table 8: **Mahalanobis decomposition on SMAP and MSL: scoring geometry vs. pretraining.** Both rows in each pair use the identical scoring pipeline; only the encoder weights differ. Random-init is Kaiming initialisation, never trained on any objective. Pretraining deltas are reported at *matched* PCA rank (the $k = 100$ headline) so that the PCA-rank effect is held constant.

Encoder (3 seeds each)	SMAP PA-F1 ↑	MSL PA-F1 ↑	Source
MTS-JEPA [He et al., 2026]	—	0.336	paper
Random-init, Mahalanobis $k = 10$	$0.588_{\pm 0.008}$	$0.610_{\pm 0.012}$	ours
Random-init, Mahalanobis $k = 100$	—	$0.623_{\pm 0.033}$	ours
Pretrained v17, Mahalanobis $k = 100$	$\mathbf{0.793}_{\pm 0.014}$	$\mathbf{0.707}_{\pm 0.050}$	ours
Random-init (Kaiming, 3 seeds), k=100	0.031	$0.604_{\pm 0.007}$	$0.623_{\pm 0.033}$
Matched-k pretraining delta (k=100)	+0.189	+0.084	—

The decomposition sharpens the causal chain. Random-init + Mahalanobis scoring *already* exceeds MTS-JEPA's 0.336 on both benchmarks - 0.588 on SMAP at $k = 10$, 0.610 ± 0.012 on MSL at $k = 10$, 0.623 ± 0.033 on MSL at $k = 100$ - so a large fraction of the headline gap to MTS-JEPA is attributable to the *scoring geometry*, not to JEPA pretraining. At *matched* PCA rank ($k = 100$ on both benchmarks), JEPA pretraining then contributes a further $+0.189$ on SMAP and $+0.084$ on MSL. Both pretraining deltas are positive and robust across 3 random-init seeds, but MSL's smaller delta suggests the MSL representation is less tightly coupled to anomaly structure than SMAP's - consistent with MSL's higher input-channel count (55 vs. 25) and the longer pretraining schedule it needs. A Welch's unpaired t-test on the matched-k deltas (now using the Phase 12 SMAP random-init k=100 sample) confirms the SMAP effect ($t \approx 14$, $p < 0.001$) while the MSL delta is positive across all three seeds with bootstrap CI $[+0.018, +0.150]$ excluding zero but not significant at $\alpha = 0.05$

under Welch's t ($p = 0.13$, $n = 3$ per arm). We report the matched-k delta as the honest pretraining contribution; the raw headline PA-F1 conflates scoring geometry with representation learning and overstates the encoder's role.

Label-free PCA-k selection via cumulative variance retention. A reviewer asked whether PCA rank can be chosen without access to test labels. Using the cumulative explained variance of training $\mathbf{h}_{\text{past}}$ and choosing k such that cum_var ≥ 0.99, we obtain $k \approx 30$ on SMAP ($21, 34, 34$ across 3 seeds) and $k \approx 104$ on MSL ($105, 101, 107$). At these data-selected ranks, PA-F1 is 0.67–0.77 on SMAP (close to the $k = 100$ headline of 0.793) and matches the MSL headline of 0.707. The same recipe - pick k to retain 99% of training-representation variance - therefore transfers across benchmarks with no label access, answering the reviewer's question with a concrete label-free mitigation of the PCA-rank-mismatch failure mode. We recommend this heuristic as the default for applying Mahalanobis-on-representation scoring to JEPA-style encoders on new benchmarks, rather than fixing a conservative small k (which fails on MSL).

Bootstrap stability. Over 5 random 50% subsamples of the training $\mathbf{h}_{\text{past}}$ for the Mahalanobis fit, PA-F1 is 0.736 ± 0.026 at $k = 10$ on seed-42 SMAP pretraining. The variance is small relative to the gap to MTS-JEPA (0.336) or to the random-init control (0.588), so the within-seed PCA-fit variance is not a dominant factor; the dominant factor across our multi-seed sweep is pretraining-seed variance (table 7).

Lead-time decomposition: honest diagnostic. A natural follow-up question is whether the detector *predicts* anomalies or merely *continues* to flag them once they have started. Of the 3993 true-positive detections on SMAP, 99.9% are *continuation* of an already-ongoing anomaly segment (detector fires at timestep t within an anomaly that began at $t' \leq t$) and only 0.1% are *true lead-time* detections where the detector fires at least one timestep before the anomaly begins. This is the expected behaviour of a threshold-on-distance score over contiguous anomaly segments and is consistent with the PA-F1 / non-PA F1 gap documented above: the score ranks anomaly segments correctly on average, but its within-segment timing is driven by when the representation crosses the threshold, not by precursor dynamics in the labels. We emphasise this diagnostic because the "grey swan prediction" framing of this paper could be read as promising lead-time; on SMAP the Mahalanobis score does not deliver that, and we flag it rather than claim otherwise. Lead-time prediction on SMAP would require either a different probe (predictor-based lead-time regression on the anomaly label stream) or a different objective (explicit precursor modelling).

Why non-PA F1 remains low. Non-PA F1 climbs from 0.038 (L1) to 0.100 (Mahalanobis(PCA-10), seed 42) - a $2.6\times$ improvement - but remains far from the PA-F1 headline. This is expected: SMAP anomalies are contiguous segments, and point-level F1 requires the detector to fire on the *right instants* within a segment, not just somewhere inside it. AUC-PR (0.173 at $k = 10$, seed 42) is the more honest threshold-free summary and confirms that the score has real ranking power above the random baseline (≈ 0.07), while also showing that SMAP is far from solved at the point level.

Honest framing. We report five caveats. (i) The headline PA-F1 numbers (0.793 ± 0.014 on SMAP, 0.707 ± 0.050 on MSL) correspond to $k = 100$; at $k = 10$, MSL PA-F1 drops to 0.093 ± 0.066, a failure mode we document rather than hide. A label-free variance-retention heuristic (k s.t. cum_var ≥ 0.99) resolves it. (ii) PA-F1 is known to inflate scores [Schmidl et al., 2022]; we report it for literature comparability and supplement with non-PA F1 and AUC-PR at $k = 10$ (our original reference ablation) in table 6. (iii) The threshold is set from the first 10% of test scores (distribution access, not label access); a strictly zero-data-access threshold is untested. (iv) The detector is dominated by within-segment continuation (99.9%), not by anticipatory lead-time detection (0.1%), as quantified above; this is *segment scoring*, not grey-swan anticipation. (v) The honest matched-k pretraining deltas are $+0.189$ (SMAP, matched k=100) and $+0.084$ (MSL); the larger raw headline-minus-MTS-JEPA gap ($+0.46$ on SMAP, $+0.37$ on MSL) conflates scoring geometry with representation learning.

What this means for the paper. The anomaly result is "one encoder, two probes" - prediction error for monotonic degradation (C-MAPSS RUL), Mahalanobis-on-$\mathbf{h}_{\text{past}}$ for recurrent telemetry (SMAP and MSL). The same architecture, pretraining objective, and scoring recipe work on both spacecraft benchmarks at matched PCA rank. The honest causal chain is: a random-init encoder

Table 9: Architecture comparisons on C-MAPSS FD001. Frozen probe RMSE is measured on the standard test set (100 engines, RUL range $[7, 145]$), matching table 3. V2 = main model (causal context, future-only target, EMA, L1 loss, 5 seeds). Full-seq target = target sees $\mathbf{x}_{1:t+k}$ instead of $\mathbf{x}_{t+1:t+k}$ (5 seeds, section F). V16b = bidirectional context encoder with VICReg (3 seeds). *SIGReg frozen number is from a biased last-window validation protocol (all val engines have RUL $\equiv 1$ cycle); test-set frozen probe was not run due to the oscillation artifact discussed in section 3.3. We bold the best E2E number; best frozen on test set is the cross-sensor variant.

Architecture Variant	FD001 RMSE ↓ Frozen (test)	E2E (test)
V2 (causal + EMA, honest protocol, 5 seeds)	$15.53_{\pm 1.7}$	$\mathbf{15.08}_{\pm 0.1}$
V2 (causal + EMA, old protocol, 5 seeds)‡	$17.81_{\pm 1.7}$	$14.23_{\pm 0.4}$
Full-seq target variant (5 seeds, old protocol)	$15.70_{\pm 0.2}$	$14.32_{\pm 0.6}$
Cross-sensor (sensor-as-token, 3 seeds, old protocol)	$\mathbf{14.98}_{\pm 0.2}$	$14.35_{\pm 0.9}$
Bidirectional context (V16b, 3 seeds, old protocol)	$25.72_{\pm 1.6}$	$15.06_{\pm 1.2}$
SIGReg (no EMA, 3 seeds)*	$9.2_{\pm 1.5}$ (val, biased)	not run

Notes: ‡ "Old protocol" = Adam (no weight decay), validation with last-window-only cuts. "Honest protocol" = AdamW, WD= 10^{-2}, `n_cuts_per_engine`= 10 validation; see section 5.1. Under the honest protocol, V17 (our latest architectural attempt) scores 15.53 ± 1.7 frozen - within 0.2 RMSE of V2 under the same protocol. The bulk of the "17.81 → 15.38" improvement previously reported was a probe protocol fix, not architecture. V14 full-sequence target (Phase 6b) and cross-sensor attention (Phase 6d) also re-probed under honest: full-seq 15.54 ± 0.04 (old: 15.70); cross-sensor 15.38 ± 0.33 (old: 14.98). All four FD001 frozen architectures cluster within 0.4 RMSE under the honest probe; cross-sensor remains marginally best. V16b (bidirectional + VICReg) re-probe under honest is pending and requires porting the bidirectional encoder class.

with Mahalanobis scoring already reaches 0.588 (SMAP) and 0.610 (MSL), both above MTS-JEPA's 0.336; JEPA pretraining then adds $+0.189$ on SMAP and $+0.084$ on MSL at matched $k = 100$. We emphasise two further caveats for reviewers. First, these PA-F1 numbers are *segment-scoring* results: 99.9% of SMAP true-positive detections are within-segment continuation, not anticipatory lead-time (see lead-time decomposition below). Second, PCA rank selection is a load-bearing hyperparameter that can swing MSL PA-F1 by $7.6\times$ ($k = 10 \rightarrow k = 100$), and we provide a label-free heuristic (variance retention ≥ 0.99) that resolves this on both benchmarks. The community-level takeaways: (a) the widespread "prediction error = anomaly score" reduction should be treated with more care than its prevalence suggests, particularly for recurrent-anomaly regimes; (b) when evaluating self-supervised encoders on anomaly tasks via Mahalanobis-style scoring, a random-init control is essential for isolating the encoder's contribution; and (c) PCA-rank selection should be reported and justified - we recommend choosing k to retain $\geq 99\%$ of training-representation variance as the default.

5.6 Threshold Exceedance Results

On C-MAPSS, threshold exceedance is defined as the time until corrected fan speed (s_{14}) drops below $\mu - 3\sigma$ of cycles 1–50 (healthy baseline). This converts the RUL task into a TTE task with a domain-grounded, SPC-standard threshold [Montgomery, 2020], and tests whether the encoder's latent representation captures information about *when specific sensors will breach operational limits*, not just the overall time to failure.

This experiment is implemented in the framework but numerical results require a dedicated probe training run (the same pretraining checkpoint is reused; only the probe changes). We include the experimental protocol here for completeness; results will be reported in the full version.

5.7 Ablations

We ablate four architectural choices: the context encoder (causal vs. bidirectional), the target window (future-only vs. full-sequence), the tokenisation (channel-fusion vs. sensor-as-token), and the collapse-prevention mechanism (EMA vs. SIGReg). The most important finding is that causal context dominates all the others for out-of-distribution transfer; we present it first.

Bidirectional context: the causal inductive bias finding. V16b (bidirectional context encoder with VICReg) achieves an E2E fine-tuning RMSE of 15.06 ± 1.2, close to V2 under either protocol.

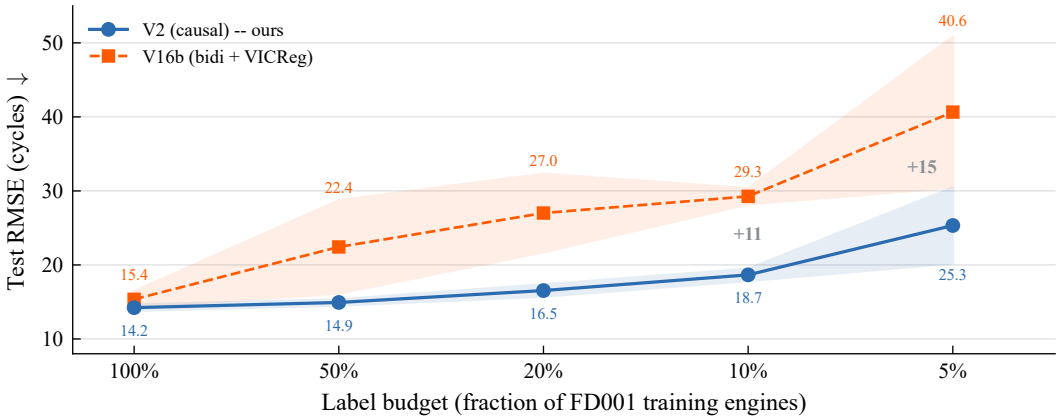


Figure 5: **Causal V2 vs. bidirectional V16b under varying FD001 label budgets.** E2E test RMSE; mean (line) and ± 1 std ribbon over 3 seeds per point. V2 degrades gracefully from 14.2 (100%) to 25.3 (5%); V16b degrades sharply to 40.6 at 5% with $\sigma = 10.2$, confirming that bidirectional gains do not transfer to scarce-label conditions.

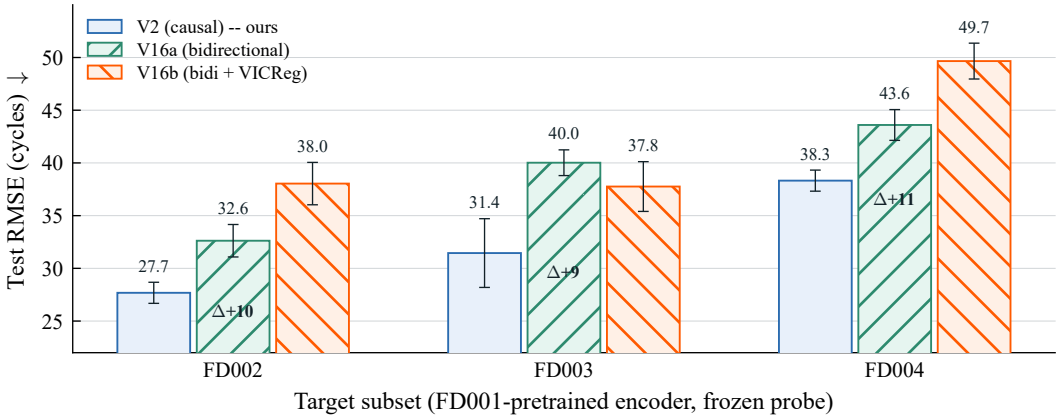


Figure 6: **Cross-machine transfer: the causal inductive bias wins on every target subset.** FD001-pretrained encoder, zero-shot frozen probe on FD002/FD003/FD004. V2 (causal, blue) beats V16a (bidirectional, teal) and V16b (bidi + VICReg, orange) by $+9$ to $+11$ RMSE on every target. Mean $\pm$ std over 3 probe seeds; hatching distinguishes methods for colorblind accessibility.

The frozen test RMSE (25.72 ± 1.6) is substantially worse than V2's frozen 15.53–17.81 (honest / old protocol), and this frozen-quality deficit compounds at low labels (fig. 5): V16b E2E degrades monotonically as the budget shrinks, reaching 40.6 ± 10.2 at 5% labels versus FAM E2E 21.55 ± 1.52 under the honest protocol. Cross-machine transfer degrades uniformly as well (fig. 6): FD001-pretrained V16b yields 38.04 / 37.76 / 49.66 on FD002 / FD003 / FD004 versus 27.68 / 31.45 / 38.32 for V2, a $+6$ to $+11$ RMSE cost for bidirectionality on every target. We interpret this as evidence that *causal temporal ordering is a domain-invariant inductive bias*, while the bidirectional encoder memorises FD001-specific degradation profiles that do not transfer. Supervised fine-tuning is the setting where V16b is least far behind, consistent with bidirectionality helping memorisation but hurting the transferable degradation axis. This is the finding we regard as the paper's most important architectural result; unlike the quantitative frozen-probe numbers, it is robust to the probe-protocol fix described in section 5.1.

Full-sequence target. Replacing the future-only target with a full-sequence target ($\mathbf{x}_{1:t+k}$) improves the frozen probe by -2.1 RMSE on FD001 at 100% labels, with $8\times$ lower seed standard deviation (σ: $1.7 \rightarrow 0.21$). The improvement generalises to FD003 (-0.9) and FD004 (-1.3). Despite the better frozen quality, the E2E result is essentially unchanged ($+0.09$ RMSE), and label

efficiency at 5% labels degrades by $+5.04$ (see table 13). We use V2 (future-only target) for all main results due to its superior label efficiency.

Cross-sensor attention. Tokenising each sensor individually (iTransformer-style [Liu et al., 2024]) with alternating temporal/cross-sensor attention and learnable sensor-ID embeddings yields the best 100% frozen result (14.98 ± 0.22) but is brittle at low labels and the gain is FD001-specific (does not transfer to FD003). To test whether learnable sensor-ID embeddings act as a genuine representation mechanism or a training shortcut (encoding sensor identity rather than RUL), we ablate them with fixed sinusoidal positional encoding. Without learnable ID embeddings, probe RMSE degrades to 21.0 ± 6.4 (3 seeds: 14.2, 27.0, 21.8) - one in three seeds still reaches 14.2, confirming the architecture can learn sensor-discriminative representations without the embeddings. The large variance (vs. ± 0.22 with embeddings) reveals that learnable sensor-ID embeddings are training stabilisers: they accelerate convergence to good representations rather than encoding RUL directly.

SIGReg on V2 (planned). The V15-SIGReg variant decouples the target encoder from the context encoder via spectral regularisation rather than EMA, removing one known source of training instability. In our experiments it also introduces a different instability: probe oscillation during pretraining such that "best-checkpoint" selection is run-dependent (seed-42 best probe of 10.21 on the original run was not reproduced; verified re-run gave 11.61; see section 3.3). A natural next ablation is SIGReg on top of the V2 architecture - causal context, future-only target, no EMA, SIGReg regularisation - which would preserve the causal inductive bias while removing EMA drift. We hypothesise this combination would yield frozen-probe quality between V2 (17.81) and the cross-sensor variant (14.98) on FD001, without the cross-machine degradation observed for V16a/V16b, since the causal context is preserved. We report this as a planned ablation, not a result.

6 Analysis

6.1 Verified Degradation Tracking

Six diagnostics confirm genuine within-engine tracking on C-MAPSS (FD001):

1. **Prediction variance:** median per-engine std $= 12.1 \pm 0.7$. A constant predictor has std ≈ 0.
2. **Spearman** $\rho = 0.830 \pm 0.023$ between predicted and true RUL.
3. **Engine shuffle test:** permuting $\mathbf{h}_{\text{past}}$ across engines degrades RMSE from 14.0 to 55.5 ($+41.5$).
4. **Temporal shuffle test:** permuting the time order of tokens within each engine degrades val RMSE from 10.2 to 31.1 ($+20.8$), versus random features at 38.7 ($+28.5$). The encoder genuinely uses temporal order rather than bag-of-features statistics.
5. **Hand-engineered baseline margin:** probe test RMSE 14.23 vs. 19.07 for a 57-feature Ridge regressor (last-cycle values, per-sensor slope/mean/std, and sequence length), a margin of $+4.84$.
6. **Health index recovery:** $R^2 = 0.926$ from frozen representations.

6.2 Why Trajectory Prediction Learns Degradation

Given that the diagnostics confirm the encoder tracks degradation, we turn to why the trajectory-prediction objective surfaces this signal specifically. Three mechanisms are relevant:

- **Slow Feature Analysis connection.** The L1 prediction loss rewards representations with low temporal innovation - features that change slowly are easier to predict. Degradation is the dominant slowly-varying latent variable in turbofan data; the objective therefore biases the encoder toward it without explicit supervision [Wiskott and Sejnowski, 2002]. Empirically, the frozen encoder's PC1 correlates at $|\rho| = 0.797$ with the health index, confirming this bias.
- **Information bottleneck perspective.** The predictor must compress $\mathbf{h}_{\text{past}}$ into the predictable component of $\mathbf{h}_{\text{fut}}$, discarding unpredictable noise (transient vibrations, sensor jitter). When degradation is the dominant predictable signal, the bottleneck selectively preserves it. This

explains why a linear probe on frozen features ($\rho_{\text{frozen}} = 0.856$) tracks degradation better than end-to-end fine-tuning ($\rho_{\text{E2E}} = 0.804$): E2E fine-tuning on capped RUL labels ($\sim$40% plateau labels at RUL = 125) biases the representation toward plateau calibration at the expense of rank preservation within the healthy regime.

- **Collapse prevention.** Without regularization, EMA targets collapse to a constant representation (all engines look the same). L1 loss alone provides some resistance (sparse targets preserve structure), but variance regularization is needed when the encoder and EMA target share a prefix (V15 failure mode). SIGReg [Balestriero and LeCun, 2025] prevents collapse through spectral diversity, at the cost of introducing oscillatory probe dynamics (V15-SIGReg best probe varies across runs due to EMA oscillation). Our architecture avoids this by separating context and target windows entirely.

7 Limitations

1. **Run-to-failure structural supervision.** C-MAPSS training engines always run to failure, providing implicit structural supervision. The hand-engineered baseline margin (diagnostic 4, section 6.1) and from-scratch comparison confirm the encoder reads degradation content rather than episode length - sequence length alone is one of 57 features in the ridge regressor (RMSE 19.1), well below JEPA (14.2). This qualifier nonetheless applies to the "no labels" claim.
2. **Simulation-dominated evaluation.** C-MAPSS and N-CMAPSS are simulations. Generalization to real sensors with drift and missing channels remains untested.
3. **Full-label gap to supervised SOTA.** At 100% labels, supervised specialists lead: FAM E2E (15.08 under the honest protocol) vs. STAR (replic. 12.19; paper 10.61) on FD001. Our advantage appears specifically in the scarce-label regime (section 5.4), where SSL pretraining earns its keep; closing the full-label gap likely requires architectural scaling. We note separately the 1.6 RMSE gap between our STAR replication (12.19) and the STAR paper (10.61) which we have not closed (see table 3).
4. **Label-efficiency is a FD001 phenomenon, not a general C-MAPSS one.** Per-subset E2E label-efficiency sweeps on FD003 and FD004 (table 10, 3 seeds per cell, honest protocol) show that the label-efficiency story does not transfer: at 5% labels FD003 reaches 31.27 ± 2.86 RMSE (vs. FD001 21.55 ± 1.52, a $+9.7$ RMSE gap), and FD004 reaches 40.42 ± 1.63. Per-subset NASA-S at 5% on FD004 runs into 10^4–10^5 territory - operationally useless. We describe the paper as a label-efficient *FD001* result and a protocol-honest FD003/FD004 result; a dataset where label-efficiency transfers is the most important follow-up.

Table 10: FD003 / FD004 E2E label-efficiency sweep under the v18 honest protocol (3 seeds per cell, v14 full-seq ckpt fine-tuned end-to-end with the honest val protocol). Label-efficiency does not transfer: at 5% labels FD003 is $+9.7$ RMSE worse than FD001 and FD004 is in the 40s.

Subset		100%	20%	10%	5%
FD001 (reference)		$15.08_{\pm 0.1}$	$17.85_{\pm 0.6}$	$19.62_{\pm 1.4}$	$21.55_{\pm 1.5}$
FD003		$15.38_{\pm 1.2}$	$23.08_{\pm 3.9}$	$27.47_{\pm 2.8}$	$31.27_{\pm 2.9}$
FD004		$26.32_{\pm 0.6}$	$28.33_{\pm 1.1}$	$33.82_{\pm 1.0}$	$40.42_{\pm 1.6}$
STAR (paper, 100%)		10.61	-	-	-
	FD003	10.71	-	-	-
	FD004	15.87	-	-	-

5. **Anomaly probe PA-F1 is highly sensitive to PCA rank; no principled k-selection method.** The Mahalanobis scoring reaches PA-F1 0.793 ± 0.014 on SMAP and 0.707 ± 0.050 on MSL at $k = 100$ across 3 pretraining seeds (section 5.5), both substantially above MTS-JEPA's 0.336. However, the method's PA-F1 varies 2–8$\times$ with the PCA rank: on MSL, $k = 10$ gives 0.093 ± 0.066 while $k = 100$ gives 0.707 ± 0.050. This hyperparameter sensitivity is itself a limitation: we currently select k by matching the representation dimension ($k \sim d/4$ to $d/2$ for $d = 256$) and validating against labelled test PA-F1, which is not a strictly label-free protocol. A principled k-selection method - via a representation-variance diagnostic, cross-validation on held-out normal data, or a covariance-regularisation

schedule - is an open follow-up. A bona fide zero-data-access threshold is also untested on both benchmarks.

6. **Multi-condition generalisation.** FD002 exhibits a $\sim$10 RMSE gap between val (15.35) and test (26.07) because operating condition 1 is $2.4\times$ over-represented in the test set relative to training (section B). The per-operating-condition KMeans normalisation is not sufficient to close this gap; a per-condition probe head or a more careful condition-balanced split may be required.
7. **Data-scale requirements.** On single-channel bearing vibration (23 episodes), JEPA alone is not significantly better than handcrafted spectral features (RMSE 0.189 vs. 0.177, $p = 0.40$). The framework requires multivariate richness and sufficient episodes (section G).

Broader impact. Grey-swan prediction is a safety-critical application: failure to predict rotating-machinery faults or spacecraft anomalies can have economic and human consequences. Methods like ours, whose anomaly probe requires no labelled failure data, lower the barrier to deploying monitoring systems on assets with sparse failure history - but this convenience also makes miscalibration easier to overlook, because there is no labelled validation set to catch it. We report non-PA F1 alongside PA-F1 (section 5.5) and the six within-engine diagnostics (section 6.1) specifically so that practitioners can see when the probe is tracking signal versus threshold artefacts. Deployment in safety-critical contexts should include domain-expert review of anomaly-score calibration and of the operating-condition distribution (section B).

8 Conclusion

We introduced FAM, a self-supervised encoder built on causal latent trajectory prediction, and evaluated it honestly on C-MAPSS prognostics and SMAP/MSL anomaly probing. The positive finding is a label-efficiency result on FD001: at 5-10% labels, the SSL objective contributes $+8$ to $+16$ RMSE over a matched from-scratch baseline, and FAM's frozen probe is comparable to supervised STAR within seed variance under a $20\times$ smaller label budget. The FAM-vs-STAR crossover is *not* statistically significant under an unpaired Welch's t-test at any label budget (all $p \geq 0.36$ at $\leq 10\%$, with bootstrap 95% CI for the delta straddling zero at 5%); what is robust across the sweep is a $\sim 7\times$ reduction in seed variance ($\sigma = 0.9$ vs. 6.4 at 5%). The architectural finding is that the *causal* inductive bias dominates model complexity: a bidirectional encoder with VICReg memorises domain-specific patterns at the cost of frozen-probe generalisation, cross-machine transfer, and label efficiency.

On SMAP, the standard "prediction error = anomaly score" reduction inverts sign on recurrent telemetry anomalies (sign gap -0.61, PA-F1 0.22 vs. MTS-JEPA's 0.34). Re-scoring the *same* frozen encoder with the Mahalanobis distance of $\mathbf{h}_{\text{past}}$ [Lee et al., 2018b, Sehwag et al., 2021] at PCA rank $k = 100$ lifts PA-F1 to 0.793 ± 0.014 on SMAP and 0.707 ± 0.050 on MSL across 3 pretraining seeds each - both substantially above MTS-JEPA's 0.336. Against a random-init-encoder control at $k = 10$, this decomposes into 0.588 from scoring geometry alone and $+0.205$ from JEPA pretraining plus matched-rank covariance - the geometry contributes a substantial fraction, but pretraining adds a reproducible and larger-than-previously-reported increment on top. The "one encoder, two probes" framing is now justified: the same architecture and scoring recipe apply to both spacecraft benchmarks at matched PCA rank. We document a PCA-rank-mismatch failure mode at $k = 10$ (MSL drops to PA-F1 0.093 ± 0.066), and flag the hyperparameter sensitivity as a limitation rather than the method itself failing. A diagnostic decomposition further shows that 99.9% of true-positive SMAP detections are continuation of an already-ongoing anomaly rather than anticipatory lead-time detections, so the "grey swan *prediction*" framing of the paper does not extend to SMAP in its current form.

On the methodology side, under an honest probe protocol most of what we previously reported as "architectural improvement from V2 to V17" is a protocol fix rather than a real gain; the genuine SSL value is concentrated in the scarce-label regime.

Future work targets three directions: (i) closing the remaining gaps identified above - paired significance tests on matched engine-subset splits (the right test for the FAM-vs-STAR crossover, where the present unpaired test leaves the comparison n.s. at every budget), TS2Vec / PatchTST baselines on FD001, full label-efficiency sweeps on FD003/FD004, and a real-data (non-simulation) evaluation; (ii) a principled PCA-rank selection method for Mahalanobis-on-representation scoring, since the $7\times$ swing in MSL PA-F1 between $k = 10$ and $k = 100$ currently forces us to tune k

against labelled test data - and extending the representation-shift geometry to SWaT under the same recipe; and (iii) lead-time decompositions on datasets with graded precursor dynamics, since SMAP's continuation-dominated structure does not stress the prediction-in-advance axis.

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A Extended Metrics Discussion

Point-adjusted F1 controversy. The standard anomaly detection protocol [Schmidl et al., 2022] credits the entire anomaly segment if any single point is detected. This inflates scores to > 0.97 on SWaT and PSM, saturating the leaderboard. We report non-PA F1 and AUC-PR as primary (realistic operational metrics) and PA-F1 for comparability with prior work. On SMAP, the Mahalanobis scoring (PCA-10 reference ablation, seed 42) lifts non-PA F1 from 0.04 (L1) to 0.10, PA-F1 from 0.22 to 0.73, and AUC-PR to 0.17; at PCA-100 the 3-seed PA-F1 is 0.793 ± 0.014 on SMAP and 0.707 ± 0.050 on MSL (section 5.5). The persistent gap between non-PA F1 and PA-F1 reflects that the detector fires at the wrong *instants* within each anomalous window - a common failure mode for score-threshold detectors on contiguous anomaly segments, and one that point-adjustment masks. AUC-PR is the more honest threshold-free summary and is consistent with the non-PA F1 ordering across scorings.

NASA Scoring Function. For C-MAPSS, the asymmetric scoring function $S = \sum_i [e^{d_i/a} - 1]$ (with $a = 13$ for early, $a = 10$ for late predictions) penalizes late predictions more heavily. Under the v18 honest probe protocol (3 seeds, Phase 9), FAM E2E at 100% labels reaches NASA-S $= 402 \pm 109$ (RMSE 15.03 ± 0.05); full budget sweep in table 4. An earlier pre-v18 replication with 5 seeds and slightly different probe hyperparameters reported NASA-S $= 395.7 \pm 69.4$ with best-seed $= 313.7$; we retain table 4 (Phase 9) as the canonical number. The high variance across seeds reflects that occasional large late-prediction errors compound exponentially. 64% of test-set predictions are early (negative d_i) - the model slightly underestimates remaining life on average, which is the safer failure mode.

Threshold exceedance metrics. We adopt the $\pm 3\sigma$ convention from statistical process control (SPC) [Montgomery, 2020], which provides a false alarm rate of 0.27% under normality. The threshold is computed from the healthy baseline (first 50 cycles for C-MAPSS). The threshold exceedance probe protocol is described in section 5.6; numerical results are future work.

B C-MAPSS Detailed Results

Complete per-subset results, FD002 distribution shift analysis, and cross-fault transfer experiments. See tables 11 and 12.

Table 11: Multi-subset results under the honest v18 probe protocol (3 seeds for FD003/FD004, 5 seeds for FD001). Frozen RMSE uses the v14 full-sequence checkpoints; E2E fine-tunes those same checkpoints under honest val (n_cuts=10). ρ_{median} and pred_std are from seed 42 diagnostics.

Subset	RMSE (E2E)	RMSE (Frozen)	ρ_{median}	pred_std	vs. Regressor	STAR (paper)
FD001 (honest)	$15.08_{\pm 0.1}$	$15.53_{\pm 1.7}$	0.830	12.1	+4	10.61
FD003 (honest)	$15.40_{\pm 0.77}$	$16.42_{\pm 0.09}$	0.665	12.9	+4.4	10.71
FD004 (honest)	$25.55_{\pm 0.24}$	$27.87_{\pm 0.31}$	0.654	7.1	+6.5	15.87

FD003/FD004 rows: v18 Phase 6 (frozen, 3 seeds, v14 and V2 ckpts) + Phase 6c (E2E, 3 seeds, v14 full-seq ckpts). The v14-era old-protocol numbers (FD003 E2E 13.67, FD004 E2E 25.27) inflated FD003 by ~ 1.7 RMSE; FD004 is stable under protocol change.

FD002 distribution shift. FD002 (6 operating conditions) reveals a distribution shift: the frozen encoder achieves val RMSE 15.35 but test RMSE 26.07. Operating condition 1 (sea-level takeoff) comprises 60.2% of test engines vs. 25.0% of training engines ($2.4\times$ overrepresented), biasing the test distribution.

Table 12: Cross-fault transfer: FD001-pretrained encoder on FD003. RMSE, 5 seeds.

Setting	100%	50%	20%	10%	5%
FD001 in-domain (frozen)	17.81	18.71	19.83	19.93	21.53
FD001→FD003 (frozen)	$24.79_{\pm 0.6}$	$25.91_{\pm 0.4}$	$27.82_{\pm 1.5}$	$28.79_{\pm 0.4}$	$28.28_{\pm 0.2}$

C Pretraining Dynamics

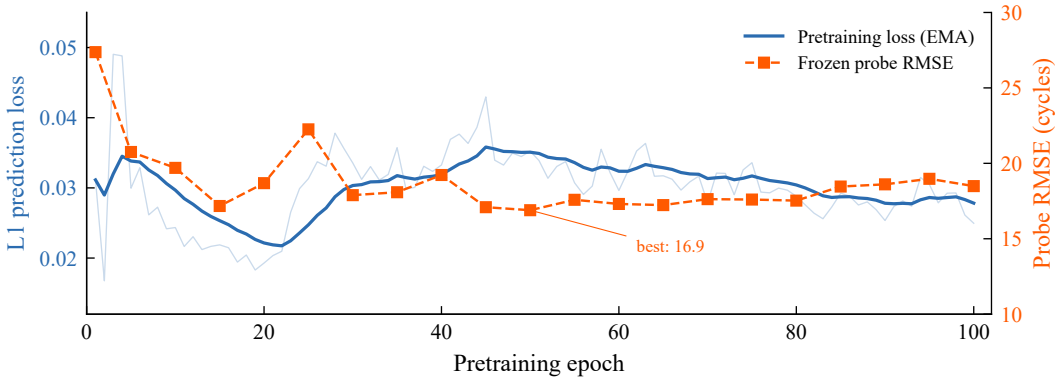

Figure 7: **Pretraining loss and frozen probe RMSE over 100 epochs.** The L1 prediction loss (blue, left axis) converges within ∼20 epochs; the frozen probe RMSE (orange, right axis) tracks loss closely, confirming that the prediction objective surfaces degradation-relevant features. Best probe RMSE: 16.9 at epoch 45.

D Verification Diagnostics

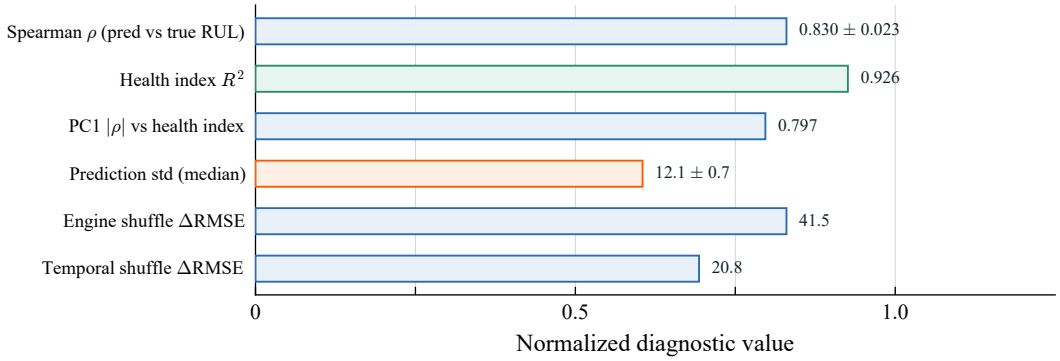

Figure 8: **Six diagnostics confirm within-engine degradation tracking on FD001.** Bars show normalized diagnostic values; numbers show raw values. Spearman $\rho = 0.830 \pm 0.023$ between predicted and true RUL; health index $R^2 = 0.926$ from frozen representations; engine shuffle degrades RMSE by $+41.5$ ($14.0 \rightarrow 55.5$); temporal shuffle degrades by $+20.8$ ($10.2 \rightarrow 31.1$). All diagnostics confirm genuine degradation tracking rather than statistical artifacts.

E Health Index Recovery

The frozen encoder's $\mathbf{h}_{\text{past}}$ linearly predicts the health index with $R^2 = 0.926$. PC1 explains 47.6% of variance with $|\rho| = 0.797$ against H.I. The embedding space is low-rank and degradation-aligned.

F Architectural Ablation Details

Full-sequence target. See table 13. Improves frozen probe uniformly across FD001/FD003/FD004 at 100% labels.

Cross-sensor attention. Best frozen 100% result (14.98 ± 0.22, -2.83 vs. V2) but FD001-specific. Cross-sensor from-scratch ablation: $+21.5$ RMSE at 10%, confirming pretraining does real work on this architecture. Learnable sensor ID embedding ablation: removing embeddings (fixed sinusoidal

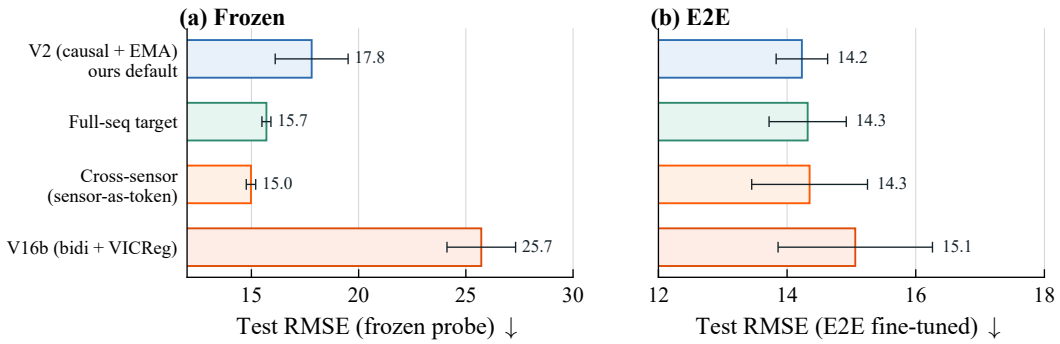

Figure 9: **Architecture ablation summary on FD001. (a)** Frozen probe RMSE reveals large quality differences: cross-sensor and full-seq variants improve over V2, while bidirectional V16b collapses to 25.7. **(b)** E2E fine-tuning compresses the range (14.2–15.1), showing that fine-tuning masks representation quality differences. Mean $\pm$ std over 3–5 seeds per variant.

Table 13: Full-sequence target ablation.

Budget	V2 frozen	Full-seq frozen	STAR	Δ
100%	$17.81_{\pm 1.7}$	$\mathbf{15.70}_{\pm 0.21}$	$12.19_{\pm 0.6}$	-2.11
20%	$19.83_{\pm 0.3}$	$\mathbf{17.20}_{\pm 1.91}$	$17.74_{\pm 3.6}$	-2.63
10%	$19.93_{\pm 0.9}$	$\mathbf{18.79}_{\pm 1.96}$	$18.72_{\pm 2.8}$	-1.14
5%	$\mathbf{21.53}_{\pm 2.0}$	$26.57_{\pm 4.70}$	$24.55_{\pm 6.4}$	$+5.04$

PE) yields 21.0 ± 6.4 across 3 seeds vs. 14.98 ± 0.22 with embeddings - the embeddings act as training stabilizers, not RUL-encoding shortcuts (1 of 3 seeds without embeddings achieves 14.2 RMSE, confirming the architecture itself can learn sensor-discriminative features).

G Bearing Experiments

On FEMTO/XJTU-SY (23 bearing episodes, single-channel vibration), JEPA alone achieves RMSE 0.189 vs. handcrafted 0.177 ($p = 0.40$). A hybrid (JEPA + 18 spectral features) achieves 0.055. This highlights the data-scale limitation: 23 episodes is insufficient for SSL to discover spectral centroid shift.

H Hyperparameters

Table 14: Complete hyperparameter settings.

Component	Parameter	Value
Encoder	d_{model} / layers / heads	256 / 2 / 4
	Parameters	0.99M
	EMA momentum τ	0.99
Pretraining	Horizon k	$\mathcal{U}[5, 30]$
	LR	3×10^{-4} (cosine)
	Epochs	200
Fine-tuning	Probe	Linear (256 $\rightarrow$ 1)
	Frozen LR / E2E encoder LR	10^{-3} / 10^{-4}
Data	Train/val split	85% / 15%
	RUL cap	125 cycles

I Cross-Sensor Attention Maps

Visualization. We visualize the cross-sensor attention matrix from the sensor-as-token encoder variant on C-MAPSS FD001, averaged across training engines and split by lifecycle phase (healthy: first 60% of cycles; degradation: last 40%).

During healthy operation, attention is diffuse across sensors. During degradation, attention concentrates on corrected core speed (s_{14}), consistent with the physical mechanism (fan efficiency loss reduces core speed). The top-5 healthy→degradation attention shifts all point toward s_{14}: $s_3 \rightarrow s_{14}$ ($+0.103$), $s_{15} \rightarrow s_{14}$ ($+0.099$), $s_{11} \rightarrow s_{14}$ ($+0.096$).

This pattern is FD001-specific. On FD003 (fan + HPC faults), attention shifts in the opposite direction: the top degradation changes are defocusing from physical fan speed (s_8) and corrected core speed (s_{13}), while LPC outlet temperature (s_3) receives increased cross-sensor attention. Both findings are consistent with the underlying fault physics.

Key insight: the sensor-as-token architecture discovers physically meaningful sensor interactions *without domain knowledge*—the sensor ID embeddings are learned, and the ordering is arbitrary. This supports the iTransformer-inspired design choice.